Tools — File Operations

Tools for reading, writing, editing, and organising files within the agent's working directory. Always visible; paths are constrained to the working-directory root.

Tool	Description
`read`	Read file contents with optional line range.
`write`	Create or overwrite a file.
`create`	Create a new file (preferred for new files).
`edit`	Replace one exact text occurrence in a file.
`multi_edit`	Apply multiple surgical edits atomically across files.
`multiedit`	Deprecated alias for `multi_edit`.
`apply_patch`	Apply a Codex-style patch (`*** Begin Patch`).
`patch`	Apply a unified diff patch.
`append_to_file`	Append text to the end of a file.
`rm`	Delete a single file.
`move_file`	Move or rename a file/directory.
`copy_file`	Copy a file to a new location.
`make_directory`	Create a directory (`mkdir -p`).
`file_info`	Return metadata for a file or directory.
`diff_files`	Unified diff between two files or strings.
`glob`	Find files matching a glob pattern.
`list`	List directory contents in a tree.
`update_file`	Alias for `write` (overwrite existing file).

Use cases: scaffolding projects, editing source, comparing files.

System instruction: "Use `edit` for single surgical replacements. `old_string` must match exactly once. Use `multi_edit` for changes across multiple files."

Fallback matching (`edit`/`multi_edit`): when byte-for-byte matching fails, a fallback cascade retries with whitespace-flexible and indent-normalised matching. A whitespace run may only fold across a line boundary when both the needle run and the matched file run contain a newline, so the flexible lane can never join or split lines.

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read

Read the contents of a text file. Files over 100 lines read without a range return the first 100 lines plus a section map and `total_lines` for planning subsequent ranged reads.

Arguments

Argument	Type	Required	Description	Typical value
`path`	string	yes	File to read	`"src/main.rs"`
`start_line`	integer	no	1-based absolute first line	`201`

Argument	Type	Required	Description	Typical value
`num_lines`	integer	no	Count of lines to read from `start_line`	`100`
`end_line`	integer	no	Absolute last line (advanced/legacy; must be >= `start_line`)	`300`

Workflow: read without a range first, then use `start_line` + `num_lines` for specific sections. Example: `start_line=201, num_lines=100` reads lines 201-300 inclusive.

Example:

```
read path="README.md"
read path="src/lib.rs" start_line=201 num_lines=100
```

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write

Write content to a file, overwriting it in full; creates parent directories as needed.

Arguments

Argument	Type	Required	Description	Typical value
`path`	string	yes	Destination path	`"src/lib.rs"`
`content`	string	yes	Complete new file contents	—

Example:

```
write path="src/new.rs" content="fn main() {}\n"
```

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create

Create a new file with the given content; creates parent directories. Truncates and overwrites if the file already exists — use `edit` for surgical changes to existing files.

Arguments

Argument	Type	Required	Description	Typical value
`path`	string	yes	Path of the file to create	`"src/module.rs"`
`content`	string	yes	Content to write	—

Example:

```
create path="docs/notes.md" content="# Notes\n"
```

---

edit

Replace exactly one occurrence of `old_string` with `new_string` in a file. Always read the target section first and copy `old_string` verbatim, including 3–5 lines of surrounding context so the needle matches exactly once. On failure nothing is written; the error includes a near-miss snippet to rebuild the needle from.

Arguments

Argument	Type	Required	Description	Typical value
`file_path`	string	yes	File to edit	`"src/main.rs"`
`old_string`	string	yes	Text to find (must match exactly once); empty string on a non-existent file creates it	—
`new_string`	string	yes	Replacement text; empty string deletes the match	—
`dry_run`	boolean	no	Resolve and preview without writing	`true`
`collapse_whitespace`	boolean	no	Fold whitespace runs; decodes `\t`, `\n`, `\r`, `\\` escapes	`false`

Example:

```
edit file_path="src/main.rs" old_string="fn main() {" new_string="fn main() -> Result<()> {"
```

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multi_edit

Apply several surgical edits across one or more files atomically. If any edit fails validation, no files are modified (all-or-nothing).

Arguments

Argument	Type	Required	Description	Typical value
`edits`	array	yes	Array of edit objects	see below

Each edit object:

Field	Type	Required	Description
`file_path`	string	yes	File to edit
`old_string`	string	yes	Text to find (exactly one match)
`new_string`	string	yes	Replacement text
`collapse_whitespace`	boolean	no	Whitespace-relaxed matching for this edit

Example:

```
multi_edit edits=[
  {"file_path":"src/a.rs","old_string":"let x = 1;","new_string":"let x = 2;"},
  {"file_path":"src/b.rs","old_string":"use a;","new_string":"use a::b;"}
]
```

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multiedit

Deprecated alias for `multi_edit`. Same parameters; prefer `multi_edit`.

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apply_patch

Apply a Codex-style patch wrapped in `*** Begin Patch` / `*** End Patch`, containing `*** Add File:`, `*** Update File:`, or `*** Delete File:` operations. All operations are validated before any file is written.

Arguments

Argument	Type	Required	Description	Typical value
`patch`	string	yes	Codex-style patch content	`"*** Begin Patch\n*** Update File: src/x.rs\n..."`
`path`	string	no	Override base directory for relative paths	`"src"`
`dry_run`	boolean	no	Validate without writing	`true`

Example:

```
apply_patch patch="*** Begin Patch\n*** Update File: src/main.rs\n@@\n fn main() {\n+    println!
```

---

patch

Apply a unified diff patch (as produced by `diff -u` or `git diff`). All hunks are validated before anything is applied.

Arguments

Argument	Type	Required	Description	Typical value
`patch`	string	yes	Unified diff content	`"--- a/src/x.rs\n+++ b/src/x.rs\n..."`
`path`	string	no	Override target file path (single-file patches)	`"src/x.rs"`
`fuzz`	integer	no	Context lines that may be dropped at hunk top/bottom	`0`

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append_to_file

Append text to the end of an existing file, creating the file and missing parent directories if needed.

Arguments

Argument	Type	Required	Description	Typical value
`path`	string	yes	File to append to	`"log/output.txt"`
`content`	string	yes	Text to append	`"done\n"`

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rm

Delete a single file. Wildcards and glob patterns are not allowed.

Arguments

Argument	Type	Required	Description
`path`	string	yes	Exact path of one file to delete

Example: `rm path="target/temp/scratch.txt"`

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move_file

Move or rename a file or directory; uses an atomic OS rename on the same filesystem. Parent directories on the destination are created if missing.

Arguments

Argument	Type	Required	Description
`source`	string	yes	Existing path
`destination`	string	yes	Target path including new name

Example: `move_file source="docs/old.md" destination="docs/new.md"`

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copy_file

Copy a single file to a new location; the source is left unchanged and parent directories are created.

Arguments

Argument	Type	Required	Description
`source`	string	yes	File to copy
`destination`	string	yes	Target path including the new file name

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make_directory

Create a directory at the given path, including missing parents (`mkdir -p`). No-op if it already exists.

Arguments

Argument	Type	Required	Description
`path`	string	yes	Directory path to create

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file_info

Return metadata for a file or directory: size in bytes, last-modified time (UTC), file type, existence, and permissions.

Arguments

Argument	Type	Required	Description
`path`	string	yes	Path to inspect

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diff_files

Show a unified diff between two files or two inline text strings. Provide either `path_a` + `path_b` or `text_a` + `text_b` — do not mix files and text for the same side.

Arguments

Argument	Type	Required	Description	Typical value
`path_a`	string	(one side)	First file (old)	`"a.rs"`
`path_b`	string	(one side)	Second file (new)	`"b.rs"`
`text_a`	string	(one side)	First inline text (old)	—
`text_b`	string	(one side)	Second inline text (new)	—
`context_lines`	integer	no	Unchanged lines shown around changes	`3`

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glob

Find files matching a glob pattern by recursively searching directories. Hidden entries and common generated directories (`node_modules`, `target`) are skipped; results are capped at 1,000 matches.

Arguments

Argument	Type	Required	Description	Typical value
`pattern`	string	yes	Glob pattern	`"**/*.rs"`, `"src/**/*.ts"`
`path`	string	no	Base directory (default: working directory)	`"crates"`

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list

List directory contents in a tree-like format, directories first.

Arguments

Argument	Type	Required	Description	Typical value
`path`	string	no	Directory to list (default: working directory)	`"src"`
`depth`	integer	no	Maximum recursion depth	`2`

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update_file

Alias for the canonical `write` tool: overwrite an existing file with new contents.

Arguments

Argument	Type	Required	Description
`path`	string	yes	File to create or overwrite
`content`	string	yes	New file contents